

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm. Examination

University Theory Examination May 2017

PH222 PHYTOCHEMISTRY-I

Date: 08/05/2017, Monday

Time: 1:30 pm to 4:30 pm

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q.1 Attempt all questions. Each question is of one mark.

20

- 1 Quercetin is the aglycone obtained from
 - A. Sennoside
 - B. Rutin
 - C. Digitoxin
 - D. Aloin
- 2 Sennosides are used as _____
 - A. Anticancer
 - B. Laxative
 - C. Cardiotonic
 - D. Antihypertensive
- 3 _____ is used in treatment of cancer
 - A. Vincristine
 - B. Rutin
 - C. Atropine
 - D. Quinine
- 4 _____ is a lysergic acid derivative
 - A. Quinine
 - B. Morphine
 - C. Ergotamine
 - D. Ephedrine
- 5 Find the alkaloid containing secondary nitrogen
 - A. Reserpine
 - B. Quinine
 - C. Ephedrine
 - D. Tubocurarine
- 6 Find the saponin structure containing drug
 - A. Glycyrrhithinic Acid
 - B. Rutin
 - C. Ephedrine
 - D. Reserpine

- 7 Morphine is obtained from _____
A. Vinca spp.
B. Cinchona spp.
C. Papaver spp.
D. Claviceps spp.
- 8 Rutin is also known as _____
A. Vitamin A
B. Vitamin P
C. Vitamin E
D. Vitamin R
- 9 One of the following is an example of Lignan
A. Rutin
B. Diosgenin
C. Podophyllotoxin
D. Sennoside
- 10 One of following is obtained from exudate of the plant
A. Morphine
B. Quinine
C. Reserpine
D. Sennoside
- 11 Application of acid in glycoside isolation is required to _____
A. Hydrolyze aglycone
B. Hydrolyze Glycone
C. Hydrolyze glycosidic Linkage
D. Solubilize glycoside
- 12 Which of the following is biosynthesized from Tyrosine
A. Morphine
B. Ergometrine
C. Diosgenin
D. Sennoside
- 13 _____ is used as central analgesic.
A. Vincristine
B. Caffeine
C. Morphine
D. Podophyllotoxin
- 14 Alkaloids are _____
A. Primary products
B. Secondary metabolites
C. Always solid in state
D. White in color
- 15 Alkaloids can be isolated as
A. Free base
B. Salts of acid
C. Both
D. None
- 16 To liberate free alkaloid as base in isolation, _____ is added
A. Potassium hydroxide
B. Sulphuric acid
C. Hydrochloric acid
D. Benzene

- 17 Pathway involved in biosynthesis of ephedrine is
 - A. Acetate melonate pathway
 - B. Acetate mevalonate pathway
 - C. Acetate glyconate pathway
 - D. Shikimic acid pathway
- 18 According to Isoprene rule, two isoprene units are joined in which manner?
 - A. Head to head
 - B. Tail to tail
 - C. Head to tail
 - D. None
- 19 Find the odd thing
 - A. Amino acids
 - B. Terpenoids
 - C. Fatty acids
 - D. Carbohydrates
- 20 If quinine is anti malarial then quinidine is ____
 - A. Anti malarial
 - B. anti viral
 - C. Anti arrhythmic
 - D. Anti pyretic

SECTION – II

Q.2 Attempt any TWO of the following:

- A Define primary and secondary metabolites of plants giving examples. Describe its inter-relationship. 05
- B Write a note on shikimic acid pathway for generation of aromatic amino acid. 05
- C Justify why Indian podophyllum is medicinally important source than American. 05
Prepare a note on Lignan obtained from podophyllum with special reference to isolation and application.

Q.3 Attempt any TWO of the following:

- A Describe in detail about biosynthesis of caffeine. 05
- B In context to isolation of quinine. 05
 - 1) Why 5% NaOH solution is added to moisten the cinchona bark powder?
 - 2) Why benzene is added in the process?
 - 3) Why sulphuric acid is added in the process?
- C Enumerate methods for basic biogenetic pathway elucidation. Write in detail about radio isotop tracer technique. 05

SECTION – III

Q.4 Attempt any FOUR of the following:

- A** Describe general procedure for isolation of Alkaloids from plant and write important steps involved in isolation with justification. **05**
- B** Write important steps involved in generation of terpenoidal compounds from acetate as precursor. **05**
- C** Write in brief about biosynthetic production of ephedrine. **05**
- D** Discuss in detail about biosynthesis of atropine. **05**
- E** Make a note on sennosides with reference to its estimation and utilization. **05**
- F** How morphine is synthesized in Opium plant? **05**

Q.5 Attempt any FOUR of the following:

- A** How biogenesis of digoxin occur in plants? **05**
 - B** Explain chemistry, isolation and estimation of glycyrrhizin. **05**
 - C** Describe sources of following: **05**
(1) Nicotine (2) Rutin (3) Reserpine (4) Morphine (5) Ergotamine
 - D** Write in detail about medicinal utilization of digitalis glucosides. **05**
 - E** Discuss stereoisomerism of natural compounds with suitable example. **05**
 - F** Describe pharmacological importance and write utilization of Ergometrine. **05**
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